# MT220 · Week 2 — Central Tendency & Dispersion

Complete study pack: notes for slides 1–49 · 100 practice questions (10 per topic) · fully worked answers. All formulas follow the lecture slides.

## Part A — Measures of Central Tendency

A **measure of central tendency** is one number that represents the whole dataset — its "middle" or most typical value. Three main measures: **Mean** (average), **Median** (middle value), **Mode** (most frequent value).

### 1. Arithmetic Mean (Slides 6–8)

| Data type | Formula | Slide example |
|---|---|---|
| Ungrouped (raw) | $\bar{X} = \Sigma X / n$ | 10, 15, 20, 25, 30 → 100/5 = **20** |
| Discrete frequency | $\bar{X} = \Sigma fX / \Sigma f$ | X 10, 20, 30; f 2, 5, 3 → 210/10 = **21** |
| Grouped (classes) | $\bar{X} = \Sigma fx / N$, x = midpoint | Running example → 745/30 = **24.83** |

- **Midpoint** x = (Lower limit + Upper limit) / 2. We assume all values of a class sit at its midpoint.
- Classes 1–10, 11–20 … are **inclusive** (they have a gap), so their true **class boundaries** are 0.5–10.5, 10.5–20.5 … Boundaries are used for median, mode and quartiles — **not** for the midpoint.
- Excel: `=AVERAGE(A1:A5)` ; frequency mean `=SUM(fx)/SUM(f)` .

**Running example (used all lecture)** — marks of 30 students:

| Class | Boundary | f | x | fx | c.f. |
|---|---|---|---|---|---|
| 1–10 | 0.5–10.5 | 4 | 5.5 | 22 | 4 |
| 11–20 | 10.5–20.5 | 6 | 15.5 | 93 | 10 |
| 21–30 | 20.5–30.5 | 10 | 25.5 | 255 | 20 |
| 31–40 | 30.5–40.5 | 8 | 35.5 | 284 | 28 |
| 41–50 | 40.5–50.5 | 2 | 45.5 | 91 | 30 |
| **Total** | | **30** | | **745** | |

Results: Mean **24.83** · Median **25.5** · Mode **27.17** · $Q_1$ **16.33** · $Q_3$ **33.63** · IQR **17.29** · QD **8.65** · $\sigma^2$ **126.22** · $\sigma$ **11.24**

### 2. Weighted Mean (Slide 9)

**$\bar{X}w = \Sigma WX / \Sigma W$** — used when values differ in importance (credit hours, quantities).
Slide: Maths 80 (4 cr), English 70 (2), Statistics 90 (3) → 730/9 = **81.1** (simple mean = 80).
If all weights are equal → weighted mean = simple mean.

### 3. Median (Slides 11–13)

- **Raw data:** sort ascending. Odd n → middle value; even n → average of the two middle values. (4, 7, 9, 10, 15 → **9**; 3, 4, 5, 8, 10, 12 → (5+8)/2 = **6.5**)
- **Discrete frequency:** find n/2, then the **first c.f. ≥ n/2**. (n = 12, n/2 = 6, first c.f. ≥ 6 is 10 at X = 30 → Median = **30**)
- **Grouped: Median = L + [(N/2 − cf) / f] × c**
  L = lower **boundary** of the median class · cf = c.f. **before** it · f = its frequency · c = class width.
  Running example: N/2 = 15 → class 21–30; Median = 20.5 + [(15 − 10)/10] × 10 = **25.5**

### 4. Mode (Slides 15–16)

- **Raw:** the most frequent value. One mode = unimodal; two+ = bimodal/multimodal; all different = no mode. Excel `=MODE.SNGL` .

- **Grouped: Mode = L + [($f_1 - f_0$) / ($2f_1 - f_0 - f_2$)] × c**
  $f_1$ = frequency of modal class (highest f), $f_0$ = class before, $f_2$ = class after (use 0 if none).
  Running example: 20.5 + [(10 − 6)/(20 − 6 − 8)] × 10 = 20.5 + 6.67 = **27.17**

## 5. Empirical Relationship (Slide 17)

**Mode = 3 × Median − 2 × Mean** (moderately skewed data only; an approximation).
Slide: Mean 20, Median 18 → Mode = 54 − 40 = **14**.

## 6. Quartiles (Slides 19–21)

Quartiles split ordered data into 4 equal parts: **$Q_1$** (25% below), **$Q_2$ = Median** (50%), **$Q_3$** (75%).

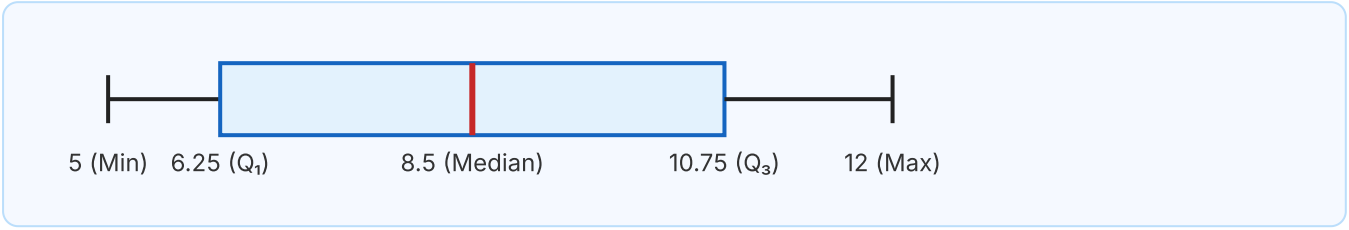
- **Raw data:** positions $Q_1$ = (n+1)/4, $Q_2$ = 2(n+1)/4, $Q_3$ = 3(n+1)/4, then interpolate.
  Slide: 5, 6, 7, 8, 9, 10, 11, 12 → $Q_1$ = 6 + 0.25(1) = **6.25**, $Q_2$ = **8.5**, $Q_3$ = **10.75**. Excel `=QUARTILE.INC(range, k)`.
- **Grouped: $Q_k = L + [(kN/4 - cf) / f] \times c$**
  Running example: $Q_1$: N/4 = 7.5 → 10.5 + [(7.5 − 4)/6] × 10 = **16.33**; $Q_3$: 3N/4 = 22.5 → 30.5 + [(22.5 − 20)/8] × 10 = **33.63**

## 7. IQR and Quartile Deviation (Slide 22)

**IQR = $Q_3 - Q_1$** — spread of the middle 50%; less affected by extreme values than the range.
**QD = IQR / 2** (semi-interquartile range).
Running example: IQR = 33.63 − 16.33 = **17.29** (17.30 with rounded Q's); QD = **8.65**.

## 8. Box Plot (Slide 23)

Five-number summary: **Min, $Q_1$, Median, $Q_3$, Max**. The box = middle 50% (IQR), line inside = median, whiskers to min and max; points beyond whiskers = possible outliers.
Slide: 5, 6, 7, 8, 9, 10, 11, 12 → Min 5, $Q_1$ 6.25, Median 8.5, $Q_3$ 10.75, Max 12.




## 9. Applications (Slide 25)

- AI: mean accuracy across cross-validation folds; **median** inference time when a few requests spike (outliers).
- Cyber: mean/median response time = performance baseline; **mode** = most common alert type in logs.

# Part B — Measures of Dispersion

**Dispersion** (variation / spread) = how far values are scattered around the mean. Same mean can hide very different spreads: Class A 45–49 (mean 47, consistent) vs Class B 20–80 (mean 50, inconsistent). Measures: Range, IQR, Variance, S.D., C.V.

## 10. Range and Coefficient of Range (Slide 28)

**Range = L − S** (largest − smallest). Quick, but uses only the two extremes.
**Coeff. of Range = (L − S) / (L + S)** — unit-free, for comparing datasets.
Slide: 150 ... 170 cm → Range = **20 cm**; Coeff. = 20/320 = **0.0625 = 6.25%**.

## 11. Variance and Standard Deviation (Slides 29–31)

| Data | Variance $\sigma^2$ | S.D. $\sigma$ |
|---|---|---|

| | | |
|---|---|---|
| Ungrouped | $\Sigma(X - \bar{X})^2 / n$ | $\sqrt{[\Sigma(X - \bar{X})^2 / n]}$ |
| Frequency (shortcut) | $\Sigma fx^2/N - (\Sigma fx/N)^2$ | $\sqrt{[\Sigma fx^2/N - (\Sigma fx/N)^2]}$ |

- Variance is in **squared units**; S.D. is back in the original units. **Smaller S.D. → more consistent.**
- Slide (ungrouped): 2, 4, 6, 8, 10 → $\Sigma(X - \bar{X})^2 = 40$ → $\sigma^2 = \mathbf{8}$, $\sigma = \mathbf{2.83}$.
- Slide (grouped): $22287.5/30 - (745/30)^2 = 742.92 - 616.69 = \mathbf{126.22}$, $\sigma = \mathbf{11.24}$.
- ⚠ $fx^2 = f \times x^2$ (e.g. $4 \times 5.5^2 = 121$) — **not** $(fx)^2$. ⚠ Check: $\Sigma(X - \bar{X})$ must equal 0.

### 12. Coefficient of Variation (Slides 32–33)

**C.V. = $(\sigma / \bar{X}) \times 100\%$** — relative spread; compares datasets with different units or means. **Lower C.V. = more consistent.** Slide Example 4.9: Price C.V. = **41.62%**, Life C.V. = **37.15%** → Price varies more, even though Life's S.D. (78.38) is much larger than Price's (7.66).

### 13. Applications of Dispersion (Slide 35)

- AI: low variance across folds = stable model; high variance = overfitting (bias–variance trade-off).
- Cyber: **3-sigma rule** — flag values beyond mean ± 3σ; a DDoS attack sharply increases variance of request rates.

## Part C — Skewness and Kurtosis

### 14. Skewness (Slides 37–38)

Skewness = **lack of symmetry**.

| Type | Tail | Order | Sk |
|---|---|---|---|
| Symmetric | — | Mean = Median = Mode | 0 |
| Positive (right) | long right tail | Mean > Median > Mode | > 0 |
| Negative (left) | long left tail | Mean < Median < Mode | < 0 |

Karl Pearson: **$Sk = (Mean - Mode) / \sigma$**, or **$Sk = 3(Mean - Median) / \sigma$** when the mode is unstable.
Running example: $(24.83 - 27.17)/11.24 = \mathbf{-0.21}$; $3(24.83 - 25.5)/11.24 = \mathbf{-0.18}$ → slightly negatively skewed.

### 15. Kurtosis (Slides 39–40)

Kurtosis = peakedness / tail-heaviness compared with Normal. **$\beta_2 = [\Sigma f(X - \bar{X})^4/N] \div [\Sigma f(X - \bar{X})^2/N]^2$** = $m_4 / m_2^2$.

| Type | $\beta_2$ | Excess ($\beta_2 - 3$) | Shape |
|---|---|---|---|
| Mesokurtic | = 3 | = 0 | Normal |
| Leptokurtic | > 3 | > 0 | Peaked, fat tails, more outliers |
| Platykurtic | < 3 | < 0 | Flat, fewer/milder outliers |

⚠ Excel `=KURT` gives **excess** kurtosis — compare with 0, not 3.

### 16. Applications (Slide 41)

- Right-skewed features (income, file size) → log-transform before training.
- High kurtosis → outliers more frequent than Normal predicts (fraud detection uses it).
- Response times are right-skewed — assuming Normal can create false anomalies.

## Part D — Excel (Slides 43–45)

| Measure | Function | Slide result (12 marks) |
|---|---|---|
| Mean / Median / Mode | =AVERAGE · =MEDIAN · =MODE.SNGL | 61.08 · 60 · 60 |

| | | |
|---|---|---|
| Range | =MAX − MIN | 82 − 45 = 37 |
| $Q_1$ / $Q_3$ | =QUARTILE.INC(range,1 / 3) | 52.75 / 68.75 |
| Sample variance / S.D. | =VAR.S · =STDEV.S (÷ n − 1) | 117.36 · 10.83 |
| Population variance / S.D. | =VAR.P · =STDEV.P (÷ n) | 107.58 · 10.37 |
| C.V. | =STDEV.S(...)/AVERAGE(...)*100 | 17.73% |
| Skewness / Excess kurtosis | =SKEW · =KURT | 0.45 · −0.14 |

Sample (part of the data) → divide by **n − 1**. Population (all of the data) → divide by **n**. Hand formulas in this lecture use **n**.

## Formula Sheet

| | |
|---|---|
| Mean | $\bar{X} = \Sigma X/n \cdot \Sigma fX/\Sigma f \cdot \Sigma fx/N$ |
| Weighted mean | $\bar{X}w = \Sigma WX / \Sigma W$ |
| Median (grouped) | $L + [(N/2 - cf)/f] \times c$ |
| Mode (grouped) | $L + [(f_1 - f_o)/(2f_1 - f_o - f_2)] \times c$ |
| Empirical | Mode = 3 Median − 2 Mean |
| Quartiles | raw: position $k(n+1)/4$ · grouped: $L + [(kN/4 - cf)/f] \times c$ |
| IQR / QD | $Q_3 - Q_1 \cdot (Q_3 - Q_1)/2$ |
| Range / Coeff. | $L - S \cdot (L - S)/(L + S)$ |
| Variance | $\Sigma(X - \bar{X})^2/n \cdot \Sigma fx^2/N - (\Sigma fx/N)^2$ |
| C.V. | $(\sigma / \bar{X}) \times 100\%$ |
| Skewness | $(\text{Mean} - \text{Mode})/\sigma \cdot 3(\text{Mean} - \text{Median})/\sigma$ |
| Kurtosis | $\beta_2 = m_4/m_2^2 \cdot \text{excess} = \beta_2 - 3$ |

## Top Mistakes to Avoid

1. Using class **limits** instead of **boundaries** for L in median/mode/quartiles.
2. Using the c.f. **of** the median class instead of the c.f. **before** it.
3. Forgetting to **sort** raw data before finding the median or quartiles.
4. Writing $(fx)^2$ instead of $f \times x^2$.
5. Forgetting the square root (giving the variance as the S.D.).
6. Comparing S.D.s of datasets with different means/units — use **C.V.**
7. Comparing Excel KURT with 3 — it is already excess kurtosis (compare with 0).
8. Using $f_o$ or $f_2$ from the wrong classes in the mode formula (0 if no class exists).

# Practice Questions (100)

Try every question on paper first, then check the worked answers in the next section. Questions marked **(Slide ...)** are taken directly from the lecture.

## Set 1: Arithmetic Mean

1. Find the mean of: 12, 15, 18, 20, 25.
2. Find the mean of: 45, 52, 38, 61, 49, 55.
3. Daily temperatures (°C) for a week are: 28, 31, 30, 29, 32, 33, 27.
4. Find the mean: X = 5, 10, 15, 20 with f = 3, 7, 6, 4.
5. Number of children per family: X = 0, 1, 2, 3, 4 with f = 4, 8, 10, 5, 3. Find the mean.
6. **(Slide Q2)** Find the mean: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.
7. Find the mean: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.
8. Find the mean: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
9. The mean of 6 numbers is 15. Five of them are 10, 12, 14, 18, 20. Find the sixth number.
10. Section A (30 students) has mean 60; Section B (20 students) has mean 70. Find the combined mean.

## Set 2: Weighted Mean

1. **(Slide Q4)** Scores 85, 90, 78 in tests worth 2, 3, 1 credit hours. Find the weighted mean.
2. Marks 75, 82, 68, 90 in subjects of 3, 4, 2, 3 credit hours. Find the weighted mean.
3. A shop sells rice at Rs 50/kg (10 kg), Rs 80/kg (5 kg), Rs 120/kg (2 kg). Find the average price per kg.
4. Grade points 4.0, 3.3, 2.7, 3.7 for courses of 3, 3, 2, 4 credit hours. Find the GPA.
5. Quiz = 20%, Mid = 30%, Final = 50%. Scores: 80, 70, 85. Find the overall score.
6. Values 60, 70, 80 all have weight 2. Find the weighted mean and compare with the simple mean.
7. Marks 80 (weight 2) and 60 (weight w) give a weighted mean of 70. Find w.
8. Values 90 (weight 1) and 60 (weight 4). Find the simple mean and the weighted mean.
9. **(AI)** A model scores 92% accuracy on 1000 test samples and 85% on 4000 samples. Find the overall accuracy.
10. **(Cyber)** Server A: average response 100 ms over 200 requests; Server B: 150 ms over 300 requests. Find the overall average response time.

## Set 3: Median

1. Find the median of: 7, 3, 9, 1, 5.
2. Find the median of: 12, 5, 8, 20, 15, 10.
3. Find the median of: 23, 19, 31, 27, 19, 25, 30.
4. Response times (ms): 100, 105, 98, 250, 102, 101. Find the median. Why is it better than the mean here?
5. Find the median: X = 1, 2, 3, 4, 5; f = 3, 5, 8, 4, 2.
6. Find the median: X = 10, 20, 30, 40, 50; f = 4, 6, 5, 3, 2.
7. **(Slide Q2)** Find the median: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.
8. Find the median: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2. (No gaps, so boundaries = limits.)
9. Find the median: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
10. Find the median: Class 100–119, 120–139, 140–159, 160–179, 180–199; f = 8, 14, 20, 12, 6.

## Set 4: Mode and the Empirical Relationship

1. Find the mode: 2, 3, 3, 5, 7, 3, 8.
2. Find the mode: 4, 6, 6, 8, 9, 9, 10.
3. Find the mode: 1, 2, 3, 4, 5.
4. Shoe sizes sold: size 6, 7, 8, 9, 10 with f = 5, 12, 20, 15, 8. Find the mode.
5. **(Slide Q2)** Find the mode: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

6. Find the mode: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.
7. Find the mode: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
8. Find the mode: Class 0–10, 10–20, 20–30, 30–40; f = 12, 8, 5, 3.
9. Mean = 50, Median = 48. Estimate the mode.
10. Mode = 30, Mean = 36. Estimate the median.

## Set 5: Quartiles, IQR, QD and Box Plot

1. Find $Q_1$, $Q_2$, $Q_3$, IQR and QD: 3, 7, 8, 12, 13, 14, 18, 21.
2. Find $Q_1$, $Q_2$, $Q_3$: 2, 4, 6, 8, 10, 12, 14.
3. Find $Q_1$, $Q_2$, $Q_3$ and IQR: 15, 9, 22, 11, 18, 25, 13, 20, 17, 10, 14.
4. **(Slide Q3)** Find $Q_1$, $Q_3$, IQR and QD: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.
5. Find $Q_1$, $Q_3$ and IQR: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.
6. Find $Q_1$, $Q_3$, IQR and QD: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
7. $Q_1 = 40$ and $Q_3 = 64$. Find the IQR and the Quartile Deviation.
8. Give the five-number summary for 3, 7, 8, 12, 13, 14, 18, 21 and describe its box plot.
9. Section A: $Q_1 = 50$, $Q_3 = 70$. Section B: $Q_1 = 45$, $Q_3 = 80$. Which section's middle 50% is more spread out?
10. For 2, 4, 6, 8, 10, 12, 14 show that $Q_2$ equals the median.

## Set 6: Range and Coefficient of Range

1. **(Slide Q5)** Find the range and coefficient of range: 12, 18, 9, 25, 30.
2. Find the range and coefficient of range: 40, 55, 32, 70, 48.
3. Temperatures (°C): 18, 25, 30, 22, 35. Find the range and coefficient of range.
4. Prices range from Rs 250 to Rs 400. Find the coefficient of range.
5. Exam marks: lowest 33, highest 97. Find the range and coefficient of range.
6. The largest value is 95 and the coefficient of range is 0.25. Find the smallest value.
7. Response times (ms): 120, 135, 128, 410, 125. Find the range. What limitation does this show?
8. A: 10, 10, 10, 10, 50 and B: 10, 20, 30, 40, 50. Find both ranges. What does this show?
9. **(Slide 27 data)** Class A: 45, 46, 47, 48, 49; Class B: 20, 35, 50, 65, 80. Compare using coefficient of range.
10. Heights: 150–190 cm. Weights: 45–90 kg. Which varies more? (Use coefficient of range.)

## Set 7: Variance and Standard Deviation

1. **(Slide Q6)** Find the variance and S.D.: 3, 6, 9, 12, 15.
2. Find the variance and S.D.: 10, 20, 30, 40, 50.
3. Find the variance and S.D.: 5, 7, 9, 11, 13, 15.
4. **(Slide 27, Class A)** Find the variance and S.D.: 45, 46, 47, 48, 49.
5. **(Slide 27, Class B)** Find the variance and S.D.: 20, 35, 50, 65, 80.
6. Find the variance and S.D.: X = 1, 2, 3, 4, 5; f = 4, 8, 10, 5, 3.
7. Find the variance and S.D.: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.
8. Find the variance and S.D.: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.
9. Find the variance and S.D.: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.
10. For 4, 8, 6, 2, 10 find the population variance (÷n) and the sample variance (÷(n − 1)).

## Set 8: Coefficient of Variation

1. **(Slide Q7)** Machines A and B: mean 10 mm; S.D. A = 0.5 mm, B = 0.8 mm. Which is more consistent?
2. X: mean 40, S.D. 8. Y: mean 120, S.D. 18. Which is more consistent?
3. Heights: mean 160 cm, S.D. 8. Weights: mean 60 kg, S.D. 6. Which varies more?
4. Batsman A scores 45, 50, 55, 60, 40; Batsman B scores 20, 90, 35, 80, 25. Who is more consistent?
5. **(Slide Example 4.9)** Prices: 8, 13, 18, 23, 30. Life (hrs): 130, 150, 180, 250, 345. Which has greater relative variation?
6. Using your answers for Class 1–10 ... 31–40 (f = 3, 5, 7, 3), find the C.V.

7. C.V. = 25% and mean = 40. Find the S.D.
8. C.V. = 12% and S.D. = 3. Find the mean.
9. **(AI)** Model A: mean accuracy 90%, S.D. 2. Model B: mean 85%, S.D. 1.5 (across folds). Which is more stable?
10. **(Slide 27 data)** Class A: mean 47, $\sigma = 1.41$; Class B: mean 50, $\sigma = 21.21$. Compare consistency.

## Set 9: Skewness

1. **(Slide Q9)** Mean = 32, Median = 30, Mode = 26, S.D. = 8. Find Sk (using mode). Type?
2. Mean = 45, Median = 48, Mode = 50, S.D. = 10. Find Sk by both formulas.
3. Mean = Median = Mode = 40, S.D. = 5. Find Sk.
4. Mean = 60, Median = 55, S.D. = 15 (mode not given). Find Sk.
5. **(Slide example)** Mean = 24.83, Median = 25.5, Mode = 27.17, S.D. = 11.24. Find Sk by both formulas.
6. Mean = 70, Median = 66, S.D. = 12. Estimate the mode, then find Sk.
7. Sk = 0.5, Mean = 40, S.D. = 8. Find the mode.
8. Mean = 50, Median = 52, Mode = 55. Without calculating, state the skewness.
9. Data: 2, 3, 3, 3, 4, 5, 9. Find the mean, median, mode, S.D. and Sk (using mode).
10. Income data usually has a long right tail. Which is larger, the mean or the median? Why?

## Set 10: Kurtosis

1. **(Slide Q10)** Excel reports excess kurtosis = +2.5. Type? Outliers?
2. Find the kurtosis of 2, 4, 6, 8, 10 using $\beta_2 = m_4/m_2^2$.
3. Find the kurtosis of 1, 5, 5, 5, 5, 5, 9.
4. Excel KURT = −1.2. Type?
5. $\beta_2 = 3$. Type? What is the excess kurtosis?
6. $m_2 = 4$ and $m_4 = 40$. Find $\beta_2$ and the type.
7. $m_2 = 5$ and $m_4 = 100$. Find $\beta_2$ and the type.
8. Kurtosis $\beta_2 = 4.2$. Find the excess kurtosis and the type.
9. Dataset A: excess kurtosis 1.5; Dataset B: −0.8. Which has fatter tails?
10. **(Cyber)** Login-attempt data has kurtosis $\beta_2 = 7$. What does this suggest?

# Worked Answers

## Set 1: Arithmetic Mean

**1.1** Find the mean of: 12, 15, 18, 20, 25.

$\Sigma X = 12 + 15 + 18 + 20 + 25 = 90$
$n = 5$
$\bar{X} = \Sigma X / n = 90 / 5 =$ **18**

**1.2** Find the mean of: 45, 52, 38, 61, 49, 55.

$\Sigma X = 45 + 52 + 38 + 61 + 49 + 55 = 300$
$n = 6$
$\bar{X} = \Sigma X / n = 300 / 6 =$ **50**

**1.3** Daily temperatures (°C) for a week are: 28, 31, 30, 29, 32, 33, 27.

$\Sigma X = 28 + 31 + 30 + 29 + 32 + 33 + 27 = 210$
$n = 7$
$\bar{X} = \Sigma X / n = 210 / 7 =$ **30**

**1.4** Find the mean: X = 5, 10, 15, 20 with f = 3, 7, 6, 4.

| X | f | fX |
|---|---|---|
| 5 | 3 | 15 |
| 10 | 7 | 70 |
| 15 | 6 | 90 |
| 20 | 4 | 80 |
| **Total** | **Σf = 20** | **ΣfX = 255** |

$\bar{X} = \Sigma fX / \Sigma f = 255 / 20 =$ **12.75**

**1.5** Number of children per family: X = 0, 1, 2, 3, 4 with f = 4, 8, 10, 5, 3. Find the mean.

| X | f | fX |
|---|---|---|
| 0 | 4 | 0 |
| 1 | 8 | 8 |
| 2 | 10 | 20 |
| 3 | 5 | 15 |
| 4 | 3 | 12 |
| **Total** | **Σf = 30** | **ΣfX = 55** |

$\bar{X} = \Sigma fX / \Sigma f = 55 / 30 =$ **1.83**

**1.6 (Slide Q2)** Find the mean: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

| Class | f | x | fx |
|---|---|---|---|
| 1 – 10 | 3 | 5.5 | 16.5 |
| 11 – 20 | 5 | 15.5 | 77.5 |
| 21 – 30 | 7 | 25.5 | 178.5 |
| 31 – 40 | 3 | 35.5 | 106.5 |
| **Total** | **N = 18** | | **Σfx = 379** |

$\bar{X} = \Sigma fx / N = 379 / 18 =$ **21.06**

**1.7** Find the mean: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.

| Class | f | x | fx |
|---|---|---|---|
| 0 – 10 | 5 | 5 | 25 |
| 10 – 20 | 8 | 15 | 120 |
| 20 – 30 | 15 | 25 | 375 |
| 30 – 40 | 10 | 35 | 350 |
| 40 – 50 | 2 | 45 | 90 |
| **Total** | **N = 40** | | **Σfx = 960** |

$\bar{X} = \Sigma fx / N = 960 / 40 =$ **24**

**1.8** Find the mean: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

| Class | f | x | fx |
|---|---|---|---|
| 21 – 30 | 6 | 25.5 | 153 |
| 31 – 40 | 10 | 35.5 | 355 |
| 41 – 50 | 14 | 45.5 | 637 |
| 51 – 60 | 7 | 55.5 | 388.5 |
| 61 – 70 | 3 | 65.5 | 196.5 |
| **Total** | **N = 40** | | **Σfx = 1730** |

$\bar{X} = \Sigma fx / N = 1730 / 40 =$ **43.25**

**1.9** The mean of 6 numbers is 15. Five of them are 10, 12, 14, 18, 20. Find the sixth number.

Total of 6 numbers = $n \times \bar{X} = 6 \times 15 = 90$
Sum of the five = 10 + 12 + 14 + 18 + 20 = 74
Sixth number = 90 − 74 = **16**

**1.10** Section A (30 students) has mean 60; Section B (20 students) has mean 70. Find the combined mean.

Total marks A = 30 × 60 = 1800; B = 20 × 70 = 1400
Combined mean = (1800 + 1400) / (30 + 20) = 3200 / 50 = **64**
(This is a weighted mean, weights = number of students.)

## Set 2: Weighted Mean

**2.1 (Slide Q4)** Scores 85, 90, 78 in tests worth 2, 3, 1 credit hours. Find the weighted mean.

| Marks | Credit hrs | W×X |
|---|---|---|
| 85 | 2 | 170 |
| 90 | 3 | 270 |
| 78 | 1 | 78 |
| **Total** | **ΣW = 6** | **ΣWX = 518** |

$\bar{X}_w = \Sigma WX / \Sigma W = 518 / 6 =$ **86.33**

**2.2** Marks 75, 82, 68, 90 in subjects of 3, 4, 2, 3 credit hours. Find the weighted mean.

| Marks | Credit hrs | W×X |
|---|---|---|
| 75 | 3 | 225 |
| 82 | 4 | 328 |
| 68 | 2 | 136 |
| 90 | 3 | 270 |
| **Total** | **ΣW = 12** | **ΣWX = 959** |

$\bar{X}_w = \Sigma WX / \Sigma W = 959 / 12 =$ **79.92**

**2.3** A shop sells rice at Rs 50/kg (10 kg), Rs 80/kg (5 kg), Rs 120/kg (2 kg). Find the average price per kg.

| Price (Rs) | Qty (kg) | W×X |
|---|---|---|
| 50 | 10 | 500 |
| 80 | 5 | 400 |
| 120 | 2 | 240 |
| **Total** | **ΣW = 17** | **ΣWX = 1140** |

$\bar{X}_w = \Sigma WX / \Sigma W = 1140 / 17 =$ **67.06**

**2.4** Grade points 4.0, 3.3, 2.7, 3.7 for courses of 3, 3, 2, 4 credit hours. Find the GPA.

| Grade pts | Credit hrs | W×X |
|---|---|---|
| 4.0 | 3 | 12 |
| 3.3 | 3 | 9.9 |
| 2.7 | 2 | 5.4 |
| 3.7 | 4 | 14.8 |
| **Total** | **ΣW = 12** | **ΣWX = 42.1** |

$\bar{X}_w = \Sigma WX / \Sigma W = 42.1 / 12 =$ **3.51**

**2.5** Quiz = 20%, Mid = 30%, Final = 50%. Scores: 80, 70, 85. Find the overall score.

| Score | Weight % | W×X |
|---|---|---|
| 80 | 20 | 1600 |
| 70 | 30 | 2100 |
| 85 | 50 | 4250 |
| **Total** | **ΣW = 100** | **ΣWX = 7950** |

$\bar{X}_w = \Sigma WX / \Sigma W = 7950 / 100 =$ **79.5**

**2.6** Values 60, 70, 80 all have weight 2. Find the weighted mean and compare with the simple mean.

| X | W | W×X |
|---|---|---|
| 60 | 2 | 120 |
| 70 | 2 | 140 |
| 80 | 2 | 160 |
| **Total** | **ΣW = 6** | **ΣWX = 420** |

$\bar{X}_w = \Sigma WX / \Sigma W = 420 / 6 =$ **70**
All weights equal → weighted mean = simple mean (60+70+80)/3 = 70 ✓

**2.7** Marks 80 (weight 2) and 60 (weight w) give a weighted mean of 70. Find w.

(80×2 + 60w) / (2 + w) = 70
160 + 60w = 140 + 70w
20 = 10w → **w = 2**

**2.8** Values 90 (weight 1) and 60 (weight 4). Find the simple mean and the weighted mean.

Simple mean = (90 + 60)/2 = 75

| X | W | W×X |
|---|---|---|
| 90 | 1 | 90 |
| 60 | 4 | 240 |
| **Total** | **ΣW = 5** | **ΣWX = 330** |

$\bar{X}_w = \Sigma WX / \Sigma W = 330 / 5 =$ **66**
Weighted mean is pulled toward 60 because it has the larger weight.

**2.9 (AI)** A model scores 92% accuracy on 1000 test samples and 85% on 4000 samples. Find the overall accuracy.

| Accuracy % | Samples | W×X |
|---|---|---|
| 92 | 1000 | 92000 |
| 85 | 4000 | 340000 |
| **Total** | **ΣW = 5000** | **ΣWX = 432000** |

$\bar{X}_w = \Sigma WX / \Sigma W = 432000 / 5000 =$ **86.4**

**2.10 (Cyber)** Server A: average response 100 ms over 200 requests; Server B: 150 ms over 300 requests. Find the overall average response time.

| Avg time (ms) | Requests | W×X |
|---|---|---|
| 100 | 200 | 20000 |
| 150 | 300 | 45000 |
| **Total** | **ΣW = 500** | **ΣWX = 65000** |

$\bar{X}_w = \Sigma WX / \Sigma W = 65000 / 500 =$ **130**

## Set 3: Median

**3.1** Find the median of: 7, 3, 9, 1, 5.

Sorted: 1, 3, 5, 7, 9
n = 5 (odd) → middle (3th) value = **5**

**3.2** Find the median of: 12, 5, 8, 20, 15, 10.

Sorted: 5, 8, 10, 12, 15, 20
n = 6 (even) → (10 + 12) / 2 = **11**

**3.3** Find the median of: 23, 19, 31, 27, 19, 25, 30.

Sorted: 19, 19, 23, 25, 27, 30, 31
n = 7 (odd) → middle (4th) value = **25**

**3.4** Response times (ms): 100, 105, 98, 250, 102, 101. Find the median. Why is it better than the mean here?

Sorted: 98, 100, 101, 102, 105, 250
n = 6 (even) → (101 + 102) / 2 = **101.5**
Mean = 756/6 = 126 → the outlier 250 pulls the mean up; the median (101.5) better represents a typical day.

**3.5** Find the median: X = 1, 2, 3, 4, 5; f = 3, 5, 8, 4, 2.

| X | f | c.f. |
|---|---|---|
| 1 | 3 | 3 |
| 2 | 5 | 8 |
| 3 | 8 | 16 |
| 4 | 4 | 20 |
| 5 | 2 | 22 |

n = 22 → n/2 = 11
First c.f. ≥ 11 is 16 (at X = 3) → Median = **3**

**3.6** Find the median: X = 10, 20, 30, 40, 50; f = 4, 6, 5, 3, 2.

| X | f | c.f. |
|---|---|---|
| 10 | 4 | 4 |
| 20 | 6 | 10 |
| 30 | 5 | 15 |
| 40 | 3 | 18 |
| 50 | 2 | 20 |

n = 20 → n/2 = 10
First c.f. ≥ 10 is 10 (at X = 20) → Median = **20**

**3.7 (Slide Q2)** Find the median: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

| Class | Boundary | f | c.f. |
|---|---|---|---|
| 1 – 10 | 0.5 – 10.5 | 3 | 3 |
| 11 – 20 | 10.5 – 20.5 | 5 | 8 |
| 21 – 30 | 20.5 – 30.5 | 7 | 15 |
| 31 – 40 | 30.5 – 40.5 | 3 | 18 |
| **Total** | | **N = 18** | |

Median: position = N/2 = 9 → class 21 – 30 (boundary 20.5 – 30.5)
L = 20.5, cf = 8, f = 7, c = 10
Median = 20.5 + [(9 − 8) / 7] × 10 = 20.5 + 1.43 = **21.93**

**3.8** Find the median: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2. (No gaps, so boundaries = limits.)

| Class | Boundary | f | c.f. |
|---|---|---|---|
| 0 – 10 | 0 – 10 | 5 | 5 |
| 10 – 20 | 10 – 20 | 8 | 13 |
| 20 – 30 | 20 – 30 | 15 | 28 |
| 30 – 40 | 30 – 40 | 10 | 38 |
| 40 – 50 | 40 – 50 | 2 | 40 |
| **Total** | | **N = 40** | |

Median: position = N/2 = 20 → class 20 – 30 (boundary 20 – 30)
L = 20, cf = 13, f = 15, c = 10
Median = 20 + [(20 − 13) / 15] × 10 = 20 + 4.67 = **24.67**

**3.9** Find the median: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

| Class | Boundary | f | c.f. |
|---|---|---|---|
| 21 – 30 | 20.5 – 30.5 | 6 | 6 |
| 31 – 40 | 30.5 – 40.5 | 10 | 16 |
| 41 – 50 | 40.5 – 50.5 | 14 | 30 |
| 51 – 60 | 50.5 – 60.5 | 7 | 37 |
| 61 – 70 | 60.5 – 70.5 | 3 | 40 |
| **Total** | | **N = 40** | |

Median: position = N/2 = 20 → class 41 – 50 (boundary 40.5 – 50.5)
L = 40.5, cf = 16, f = 14, c = 10
Median = 40.5 + [(20 − 16) / 14] × 10 = 40.5 + 2.86 = **43.36**

**3.10** Find the median: Class 100–119, 120–139, 140–159, 160–179, 180–199; f = 8, 14, 20, 12, 6.

| Class | Boundary | f | c.f. |
|---|---|---|---|
| 100 – 119 | 99.5 – 119.5 | 8 | 8 |
| 120 – 139 | 119.5 – 139.5 | 14 | 22 |
| 140 – 159 | 139.5 – 159.5 | 20 | 42 |
| 160 – 179 | 159.5 – 179.5 | 12 | 54 |
| 180 – 199 | 179.5 – 199.5 | 6 | 60 |
| **Total** | | **N = 60** | |

Median: position = N/2 = 30 → class 140 – 159 (boundary 139.5 – 159.5)
L = 139.5, cf = 22, f = 20, c = 20
Median = 139.5 + [(30 − 22) / 20] × 20 = 139.5 + 8 = **147.5**

## Set 4: Mode and the Empirical Relationship

**4.1** Find the mode: 2, 3, 3, 5, 7, 3, 8.

3 appears 3 times (most often) → **Mode = 3** (unimodal)

**4.2** Find the mode: 4, 6, 6, 8, 9, 9, 10.

6 and 9 each appear twice → **Modes = 6 and 9** (bimodal)

**4.3** Find the mode: 1, 2, 3, 4, 5.

Every value appears once → **No mode**

**4.4** Shoe sizes sold: size 6, 7, 8, 9, 10 with f = 5, 12, 20, 15, 8. Find the mode.

Highest frequency = 20 at size 8 → **Mode = 8**

**4.5 (Slide Q2)** Find the mode: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

| Class | Boundary | f |
|---|---|---|
| 1 – 10 | 0.5 – 10.5 | 3 |
| 11 – 20 | 10.5 – 20.5 | 5 |
| 21 – 30 | 20.5 – 30.5 | 7 |
| 31 – 40 | 30.5 – 40.5 | 3 |
| **Total** | | **N = 18** |

Modal class = 21 – 30 (highest f = 7); boundary 20.5 – 30.5
L = 20.5, $f_1$ = 7, $f_o$ = 5, $f_2$ = 3, c = 10
Mode = 20.5 + [(7 − 5) / (2×7 − 5 − 3)] × 10 = 20.5 + [2/6] × 10 = **23.83**

**4.6** Find the mode: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.

| Class | Boundary | f |
|---|---|---|
| 0 – 10 | 0 – 10 | 5 |
| 10 – 20 | 10 – 20 | 8 |
| 20 – 30 | 20 – 30 | 15 |
| 30 – 40 | 30 – 40 | 10 |
| 40 – 50 | 40 – 50 | 2 |
| **Total** | | **N = 40** |

Modal class = 20 – 30 (highest f = 15); boundary 20 – 30
L = 20, $f_1$ = 15, $f_o$ = 8, $f_2$ = 10, c = 10
Mode = 20 + [(15 − 8) / (2×15 − 8 − 10)] × 10 = 20 + [7/12] × 10 = **25.83**

**4.7** Find the mode: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

| Class | Boundary | f |
|---|---|---|
| 21 – 30 | 20.5 – 30.5 | 6 |
| 31 – 40 | 30.5 – 40.5 | 10 |
| 41 – 50 | 40.5 – 50.5 | 14 |
| 51 – 60 | 50.5 – 60.5 | 7 |
| 61 – 70 | 60.5 – 70.5 | 3 |
| **Total** | | **N = 40** |

Modal class = 41 – 50 (highest f = 14); boundary 40.5 – 50.5
L = 40.5, $f_1$ = 14, $f_o$ = 10, $f_2$ = 7, c = 10
Mode = 40.5 + [(14 − 10) / (2×14 − 10 − 7)] × 10 = 40.5 + [4/11] × 10 = **44.14**

**4.8** Find the mode: Class 0–10, 10–20, 20–30, 30–40; f = 12, 8, 5, 3.

| Class | Boundary | f |
|---|---|---|
| 0 – 10 | 0 – 10 | 12 |
| 10 – 20 | 10 – 20 | 8 |
| 20 – 30 | 20 – 30 | 5 |
| 30 – 40 | 30 – 40 | 3 |
| **Total** | | **N = 28** |

Modal class = 0 – 10 (highest f = 12); boundary 0 – 10
L = 0, $f_1$ = 12, $f_o$ = 0, $f_2$ = 8, c = 10
Mode = 0 + [(12 − 0) / (2×12 − 0 − 8)] × 10 = 0 + [12/16] × 10 = **7.5**
Note: the modal class is the FIRST class, so there is no class before it → $f_o$ = 0.

**4.9** Mean = 50, Median = 48. Estimate the mode.

Mode = 3 × Median − 2 × Mean = 3(48) − 2(50) = 144 − 100 = **44**

**4.10** Mode = 30, Mean = 36. Estimate the median.

Mode = 3 Median − 2 Mean → 30 = 3 Median − 72
3 Median = 102 → **Median = 34**

## Set 5: Quartiles, IQR, QD and Box Plot

**5.1** Find $Q_1$, $Q_2$, $Q_3$, IQR and QD: 3, 7, 8, 12, 13, 14, 18, 21.

Sorted: 3, 7, 8, 12, 13, 14, 18, 21 (n = 8)
$Q_1$: position 1(n+1)/4 = 2.25 → between 2th (7) and 3th (8) → 7 + 0.25(8 − 7) = **7.25**
$Q_2$: position 2(n+1)/4 = 4.5 → between 4th (12) and 5th (13) → 12 + 0.5(13 − 12) = **12.5**
$Q_3$: position 3(n+1)/4 = 6.75 → between 6th (14) and 7th (18) → 14 + 0.75(18 − 14) = **17**
IQR = 17 − 7.25 = **9.75**; QD = **4.88**

**5.2** Find $Q_1$, $Q_2$, $Q_3$: 2, 4, 6, 8, 10, 12, 14.

Sorted: 2, 4, 6, 8, 10, 12, 14 (n = 7)
$Q_1$: position 1(n+1)/4 = 2 → 2th value = **4**
$Q_2$: position 2(n+1)/4 = 4 → 4th value = **8**
$Q_3$: position 3(n+1)/4 = 6 → 6th value = **12**

**5.3** Find $Q_1$, $Q_2$, $Q_3$ and IQR: 15, 9, 22, 11, 18, 25, 13, 20, 17, 10, 14.

Sorted: 9, 10, 11, 13, 14, 15, 17, 18, 20, 22, 25 (n = 11)
$Q_1$: position 1(n+1)/4 = 3 → 3th value = **11**
$Q_2$: position 2(n+1)/4 = 6 → 6th value = **15**
$Q_3$: position 3(n+1)/4 = 9 → 9th value = **20**
IQR = **9**

**5.4 (Slide Q3)** Find $Q_1$, $Q_3$, IQR and QD: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

| Class | Boundary | f | c.f. |
|---|---|---|---|
| 1 – 10 | 0.5 – 10.5 | 3 | 3 |
| 11 – 20 | 10.5 – 20.5 | 5 | 8 |
| 21 – 30 | 20.5 – 30.5 | 7 | 15 |
| 31 – 40 | 30.5 – 40.5 | 3 | 18 |
| **Total** | | **N = 18** | |

$Q_1$: position = N/4 = 4.5 → class 11 – 20 (boundary 10.5 – 20.5)
L = 10.5, cf = 3, f = 5, c = 10
$Q_1 = 10.5 + [(4.5 - 3) / 5] \times 10 = 10.5 + 3 =$ **13.5**
$Q_3$: position = 3N/4 = 13.5 → class 21 – 30 (boundary 20.5 – 30.5)
L = 20.5, cf = 8, f = 7, c = 10
$Q_3 = 20.5 + [(13.5 - 8) / 7] \times 10 = 20.5 + 7.86 =$ **28.36**
IQR = 28.36 − 13.5 = **14.86**; QD = **7.43**

**5.5** Find $Q_1$, $Q_3$ and IQR: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.

| Class | f | c.f. |
|---|---|---|
| 0 – 10 | 5 | 5 |
| 10 – 20 | 8 | 13 |
| 20 – 30 | 15 | 28 |
| 30 – 40 | 10 | 38 |
| 40 – 50 | 2 | 40 |
| **Total** | **N = 40** | |

$Q_1$: position = N/4 = 10 → class 10 – 20 (boundary 10 – 20)
L = 10, cf = 5, f = 8, c = 10
$Q_1 = 10 + [(10 - 5) / 8] \times 10 = 10 + 6.25 =$ **16.25**
$Q_3$: position = 3N/4 = 30 → class 30 – 40 (boundary 30 – 40)
L = 30, cf = 28, f = 10, c = 10
$Q_3 = 30 + [(30 - 28) / 10] \times 10 = 30 + 2 =$ **32**
IQR = **15.75**

**5.6** Find $Q_1$, $Q_3$, IQR and QD: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

| Class | Boundary | f | c.f. |
|---|---|---|---|
| 21 – 30 | 20.5 – 30.5 | 6 | 6 |
| 31 – 40 | 30.5 – 40.5 | 10 | 16 |
| 41 – 50 | 40.5 – 50.5 | 14 | 30 |
| 51 – 60 | 50.5 – 60.5 | 7 | 37 |
| 61 – 70 | 60.5 – 70.5 | 3 | 40 |
| **Total** | | **N = 40** | |

$Q_1$: position = N/4 = 10 → class 31 – 40 (boundary 30.5 – 40.5)
L = 30.5, cf = 6, f = 10, c = 10
$Q_1 = 30.5 + [(10 - 6) / 10] \times 10 = 30.5 + 4 =$ **34.5**
$Q_3$: position = 3N/4 = 30 → class 41 – 50 (boundary 40.5 – 50.5)
L = 40.5, cf = 16, f = 14, c = 10
$Q_3 = 40.5 + [(30 - 16) / 14] \times 10 = 40.5 + 10 =$ **50.5**
IQR = **16**; QD = **8**

**5.7** $Q_1 = 40$ and $Q_3 = 64$. Find the IQR and the Quartile Deviation.

IQR = $Q_3 - Q_1 = 64 - 40 =$ **24**
QD = IQR / 2 = **12**

**5.8** Give the five-number summary for 3, 7, 8, 12, 13, 14, 18, 21 and describe its box plot.

Min = 3, $Q_1 = 7.25$, Median = 12.5, $Q_3 = 17$, Max = 21
Box from 7.25 to 17 with a line at 12.5; whiskers to 3 and 21.

**5.9** Section A: $Q_1 = 50$, $Q_3 = 70$. Section B: $Q_1 = 45$, $Q_3 = 80$. Which section's middle 50% is more spread out?

IQR A = 20, IQR B = 35 → **Section B** is more spread out

**5.10** For 2, 4, 6, 8, 10, 12, 14 show that $Q_2$ equals the median.

$Q_2$ position = 2(7+1)/4 = 4 → 4th value = 8
Median (n = 7, odd) = 4th value = 8 → **$Q_2$ = Median = 8** ✓

## Set 6: Range and Coefficient of Range

**6.1 (Slide Q5)** Find the range and coefficient of range: 12, 18, 9, 25, 30.

L = 30, S = 9
Range = 30 − 9 = **21**
Coeff. of Range = (30 − 9) / (30 + 9) = 21/39 = **0.5385** = 53.85%

**6.2** Find the range and coefficient of range: 40, 55, 32, 70, 48.

L = 70, S = 32
Range = 70 − 32 = **38**
Coeff. of Range = (70 − 32) / (70 + 32) = 38/102 = **0.3725** = 37.25%

**6.3** Temperatures (°C): 18, 25, 30, 22, 35. Find the range and coefficient of range.

L = 35, S = 18
Range = 35 − 18 = **17 °C**
Coeff. of Range = (35 − 18) / (35 + 18) = 17/53 = **0.3208** = 32.08%

**6.4** Prices range from Rs 250 to Rs 400. Find the coefficient of range.

Coeff. = (400 − 250)/(400 + 250) = 150/650 = **0.2308** = 23.08%

**6.5** Exam marks: lowest 33, highest 97. Find the range and coefficient of range.

Range = 97 − 33 = **64**
Coeff. = 64/130 = **0.4923** = 49.23%

**6.6** The largest value is 95 and the coefficient of range is 0.25. Find the smallest value.

(95 − S)/(95 + S) = 0.25
95 − S = 23.75 + 0.25S
71.25 = 1.25S → **S = 57**

**6.7** Response times (ms): 120, 135, 128, 410, 125. Find the range. What limitation does this show?

Range = 410 − 120 = **290 ms**
Limitation: one extreme value (410) makes the spread look huge; range ignores all values in between.

**6.8** A: 10, 10, 10, 10, 50 and B: 10, 20, 30, 40, 50. Find both ranges. What does this show?

Range A = 40, Range B = 40 → **same range**
But B is spread evenly and A is not → range uses only 2 values, so it can hide differences.

**6.9 (Slide 27 data)** Class A: 45, 46, 47, 48, 49; Class B: 20, 35, 50, 65, 80. Compare using coefficient of range.

A: (49 − 45)/(49 + 45) = 4/94 = **0.0426**
B: (80 − 20)/(80 + 20) = 60/100 = **0.6**
→ B is far more spread out.

**6.10** Heights: 150–190 cm. Weights: 45–90 kg. Which varies more? (Use coefficient of range.)

Heights: 40/340 = **0.1176**
Weights: 45/135 = **0.3333**
→ **Weights** vary more (units differ, so use the coefficient).

## Set 7: Variance and Standard Deviation

**7.1 (Slide Q6)** Find the variance and S.D.: 3, 6, 9, 12, 15.

$\bar{X} = 45 / 5 = 9$

| X | X − $\bar{X}$ | (X − $\bar{X}$)² |
|---|---|---|
| 3 | -6 | 36 |
| 6 | -3 | 9 |
| 9 | 0 | 0 |
| 12 | 3 | 9 |
| 15 | 6 | 36 |
| **Total** | **0** | **90** |

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 90 / 5 =$ **18**
$\sigma = \sqrt{18} =$ **4.24**

**7.2** Find the variance and S.D.: 10, 20, 30, 40, 50.

$\bar{X} = 150 / 5 = 30$

| X | X − $\bar{X}$ | (X − $\bar{X}$)² |
|---|---|---|
| 10 | -20 | 400 |
| 20 | -10 | 100 |
| 30 | 0 | 0 |
| 40 | 10 | 100 |
| 50 | 20 | 400 |
| **Total** | **0** | **1000** |

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 1000 / 5 =$ **200**
$\sigma = \sqrt{200} =$ **14.14**

**7.3** Find the variance and S.D.: 5, 7, 9, 11, 13, 15.

$\bar{X} = 60 / 6 = 10$

| X | X − $\bar{X}$ | (X − $\bar{X}$)² |
|---|---|---|
| 5 | -5 | 25 |
| 7 | -3 | 9 |
| 9 | -1 | 1 |
| 11 | 1 | 1 |
| 13 | 3 | 9 |
| 15 | 5 | 25 |
| **Total** | **0** | **70** |

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 70 / 6 =$ **11.67**
$\sigma = \sqrt{11.67} =$ **3.42**

**7.4 (Slide 27, Class A)** Find the variance and S.D.: 45, 46, 47, 48, 49.

$\bar{X} = 235 / 5 = 47$

| X | X − $\bar{X}$ | (X − $\bar{X}$)² |
|---|---|---|
| 45 | -2 | 4 |
| 46 | -1 | 1 |
| 47 | 0 | 0 |
| 48 | 1 | 1 |
| 49 | 2 | 4 |
| **Total** | **0** | **10** |

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 10 / 5 =$ **2**
$\sigma = \sqrt{2} =$ **1.41**

**7.5 (Slide 27, Class B)** Find the variance and S.D.: 20, 35, 50, 65, 80.

$\bar{X} = 250 / 5 = 50$

| X | X − $\bar{X}$ | (X − $\bar{X}$)² |
|---|---|---|
| 20 | -30 | 900 |
| 35 | -15 | 225 |
| 50 | 0 | 0 |
| 65 | 15 | 225 |
| 80 | 30 | 900 |
| **Total** | **0** | **2250** |

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 2250 / 5 =$ **450**
$\sigma = \sqrt{450} =$ **21.21**

**7.6** Find the variance and S.D.: X = 1, 2, 3, 4, 5; f = 4, 8, 10, 5, 3.

| X | f | fX | $fX^2$ |
|---|---|---|---|
| 1 | 4 | 4 | 4 |
| 2 | 8 | 16 | 32 |
| 3 | 10 | 30 | 90 |
| 4 | 5 | 20 | 80 |
| 5 | 3 | 15 | 75 |
| **Total** | **30** | **85** | **281** |

$\bar{X} = 85/30 = 2.83$
$\sigma^2 = \Sigma fX^2/N - (\Sigma fX/N)^2 = 281/30 - (2.83)^2 = 9.37 - 8.03 =$ **1.34**
$\sigma =$ **1.16**

**7.7** Find the variance and S.D.: Class 1–10, 11–20, 21–30, 31–40; f = 3, 5, 7, 3.

| Class | f | x | fx | $fx^2$ |
|---|---|---|---|---|
| 1 – 10 | 3 | 5.5 | 16.5 | 90.75 |
| 11 – 20 | 5 | 15.5 | 77.5 | 1201.25 |
| 21 – 30 | 7 | 25.5 | 178.5 | 4551.75 |
| 31 – 40 | 3 | 35.5 | 106.5 | 3780.75 |
| **Total** | **N = 18** | | **Σfx = 379** | **$\Sigma fx^2$ = 9624.5** |

$\sigma^2 = \Sigma fx^2/N - (\Sigma fx/N)^2 = 9624.5/18 - (379/18)^2 = 534.69 - (21.06)^2 = 534.69 - 443.34 =$ **91.36**
$\sigma = \sqrt{91.36} =$ **9.56**

**7.8** Find the variance and S.D.: Class 0–10, 10–20, 20–30, 30–40, 40–50; f = 5, 8, 15, 10, 2.

| Class | f | x | fx | $fx^2$ |
|---|---|---|---|---|
| 0 – 10 | 5 | 5 | 25 | 125 |
| 10 – 20 | 8 | 15 | 120 | 1800 |
| 20 – 30 | 15 | 25 | 375 | 9375 |
| 30 – 40 | 10 | 35 | 350 | 12250 |
| 40 – 50 | 2 | 45 | 90 | 4050 |
| **Total** | **N = 40** | | **Σfx = 960** | **$\Sigma fx^2$ = 27600** |

$\sigma^2 = \Sigma fx^2/N - (\Sigma fx/N)^2 = 27600/40 - (960/40)^2 = 690 - (24)^2 = 690 - 576 =$ **114**
$\sigma = \sqrt{114} =$ **10.68**

**7.9** Find the variance and S.D.: Class 21–30, 31–40, 41–50, 51–60, 61–70; f = 6, 10, 14, 7, 3.

| Class | f | x | fx | $fx^2$ |
|---|---|---|---|---|
| 21 – 30 | 6 | 25.5 | 153 | 3901.5 |
| 31 – 40 | 10 | 35.5 | 355 | 12602.5 |
| 41 – 50 | 14 | 45.5 | 637 | 28983.5 |
| 51 – 60 | 7 | 55.5 | 388.5 | 21561.75 |
| 61 – 70 | 3 | 65.5 | 196.5 | 12870.75 |
| **Total** | **N = 40** | | **Σfx = 1730** | **$\Sigma fx^2$ = 79920** |

$\sigma^2 = \Sigma fx^2/N - (\Sigma fx/N)^2 = 79920/40 - (1730/40)^2 = 1998 - (43.25)^2 = 1998 - 1870.56 =$ **127.44**
$\sigma = \sqrt{127.44} =$ **11.29**

**7.10** For 4, 8, 6, 2, 10 find the population variance (÷n) and the sample variance (÷(n − 1)).

$\bar{X} = 30 / 5 = 6$

| X | X − $\bar{X}$ | (X − $\bar{X}$)² |
|---|---|---|
| 4 | -2 | 4 |
| 8 | 2 | 4 |
| 6 | 0 | 0 |
| 2 | -4 | 16 |
| 10 | 4 | 16 |
| **Total** | **0** | **40** |

$\sigma^2 = \Sigma(X - \bar{X})^2 / n = 40 / 5 =$ **8**
$\sigma = \sqrt{8} =$ **2.83**
Sample variance (Excel VAR.S) = $\Sigma(X - \bar{X})^2/(n - 1) = 40/4 =$ **10**

## Set 8: Coefficient of Variation

**8.1 (Slide Q7)** Machines A and B: mean 10 mm; S.D. A = 0.5 mm, B = 0.8 mm. Which is more consistent?

C.V.(A) = 0.5/10 × 100 = **5%**
C.V.(B) = 0.8/10 × 100 = **8%**
→ **Machine A** is more consistent (lower C.V.)

**8.2** X: mean 40, S.D. 8. Y: mean 120, S.D. 18. Which is more consistent?

C.V.(X) = 8/40 × 100 = **20%**
C.V.(Y) = 18/120 × 100 = **15%**
→ **Y** is more consistent, even though its S.D. is larger

**8.3** Heights: mean 160 cm, S.D. 8. Weights: mean 60 kg, S.D. 6. Which varies more?

C.V. heights = 5%, C.V. weights = **10%** → **weights** vary more

**8.4** Batsman A scores 45, 50, 55, 60, 40; Batsman B scores 20, 90, 35, 80, 25. Who is more consistent?

A: mean = 50, σ = 7.07, C.V. = **14.14%**
B: mean = 50, σ = 29.15, C.V. = **58.31%**
→ **Batsman A** is more consistent

**8.5 (Slide Example 4.9)** Prices: 8, 13, 18, 23, 30. Life (hrs): 130, 150, 180, 250, 345. Which has greater relative variation?

Price: mean 18.4, σ 7.66, C.V. = **41.62%**
Life: mean 211, σ 78.38, C.V. = **37.15%**
→ **Price** varies more relatively (despite the smaller S.D.)

**8.6** Using your answers for Class 1–10 … 31–40 (f = 3, 5, 7, 3), find the C.V.

Mean = 21.06, σ = 9.56
C.V. = 9.56/21.06 × 100 = **45.39%**

**8.7** C.V. = 25% and mean = 40. Find the S.D.

σ = C.V. × mean / 100 = 25 × 40 / 100 = **10**

**8.8** C.V. = 12% and S.D. = 3. Find the mean.

Mean = σ × 100 / C.V. = 3 × 100 / 12 = **25**

**8.9 (AI)** Model A: mean accuracy 90%, S.D. 2. Model B: mean 85%, S.D. 1.5 (across folds). Which is more stable?

C.V.(A) = 2/90 × 100 = **2.22%**
C.V.(B) = 1.5/85 × 100 = **1.76%**
→ **Model B** is more stable

**8.10 (Slide 27 data)** Class A: mean 47, σ = 1.41; Class B: mean 50, σ = 21.21. Compare consistency.

C.V.(A) = 1.41/47 × 100 = **3%**
C.V.(B) = 21.21/50 × 100 = **42.42%**
→ **Class A** is far more consistent

## Set 9: Skewness

**9.1 (Slide Q9)** Mean = 32, Median = 30, Mode = 26, S.D. = 8. Find Sk (using mode). Type?

Sk = (32 − 26)/8 = **0.75** → **positively skewed**

**9.2** Mean = 45, Median = 48, Mode = 50, S.D. = 10. Find Sk by both formulas.

Sk = (45 − 50)/10 = **−0.5**
Sk = 3(45 − 48)/10 = **−0.9**
→ **negatively skewed**

**9.3** Mean = Median = Mode = 40, S.D. = 5. Find Sk.

Sk = (40 − 40)/5 = **0** → **symmetric**

**9.4** Mean = 60, Median = 55, S.D. = 15 (mode not given). Find Sk.

Use the median formula: Sk = 3(60 − 55)/15 = 15/15 = **1.0** → **positively skewed**

**9.5 (Slide example)** Mean = 24.83, Median = 25.5, Mode = 27.17, S.D. = 11.24. Find Sk by both formulas.

Sk = (24.83 − 27.17)/11.24 = **−0.21**
Sk = 3(24.83 − 25.5)/11.24 = **−0.18**
→ slightly **negatively skewed**

**9.6** Mean = 70, Median = 66, S.D. = 12. Estimate the mode, then find Sk.

Mode = 3(66) − 2(70) = 198 − 140 = 58
Sk = (70 − 58)/12 = **1.0** (and 3(70 − 66)/12 = 1.0 ✓) → positive

**9.7** Sk = 0.5, Mean = 40, S.D. = 8. Find the mode.

0.5 = (40 − Mode)/8 → 40 − Mode = 4 → **Mode = 36**

**9.8** Mean = 50, Median = 52, Mode = 55. Without calculating, state the skewness.

Mean < Median < Mode → **negatively (left) skewed**

**9.9** Data: 2, 3, 3, 3, 4, 5, 9. Find the mean, median, mode, S.D. and Sk (using mode).

Mean = 29/7 = 4.14, Median = 3, Mode = 3
σ = √(32.86/7) = 2.17
Sk = (4.14 − 3)/2.17 = **0.53** → **positively skewed** (tail toward 9)

**9.10** Income data usually has a long right tail. Which is larger, the mean or the median? Why?

**Mean > Median** — positive skew: a few very high incomes pull the mean to the right (the mean follows the tail).

## Set 10: Kurtosis

**10.1 (Slide Q10)** Excel reports excess kurtosis = +2.5. Type? Outliers?

Excess > 0 → **Leptokurtic** (peaked, fat tails)
→ **more extreme outliers** than a Normal distribution

**10.2** Find the kurtosis of 2, 4, 6, 8, 10 using $\beta_2 = m_4/m_2^2$.

Mean = 6; deviations −4, −2, 0, 2, 4
$m_2 = (16+4+0+4+16)/5 = 8$
$m_4 = (256+16+0+16+256)/5 = 108.8$
$\beta_2 = 108.8/8^2 = 108.8/64 =$ **1.7** < 3 → **Platykurtic** (excess = -1.3)

**10.3** Find the kurtosis of 1, 5, 5, 5, 5, 5, 9.

Mean = 5; deviations −4, 0, 0, 0, 0, 0, 4
$m_2 = 32/7 = 4.57$
$m_4 = 512/7 = 73.14$
$\beta_2 = 73.14/4.57^2 =$ **3.5** > 3 → **Leptokurtic** (peak at 5, extreme tails)

**10.4** Excel KURT = −1.2. Type?

Excess < 0 → **Platykurtic** (flat, fewer/milder outliers)

**10.5** $\beta_2 = 3$. Type? What is the excess kurtosis?

Excess = 3 − 3 = 0 → **Mesokurtic** (same as Normal)

**10.6** $m_2 = 4$ and $m_4 = 40$. Find $\beta_2$ and the type.

$\beta_2 = 40/4^2 = 40/16 =$ **2.5** < 3 → **Platykurtic** (excess −0.5)

**10.7** $m_2 = 5$ and $m_4 = 100$. Find $\beta_2$ and the type.

$\beta_2 = 100/5^2 = 100/25 =$ **4** > 3 → **Leptokurtic** (excess +1)

**10.8** Kurtosis $\beta_2 = 4.2$. Find the excess kurtosis and the type.

Excess = 4.2 − 3 = **1.2** > 0 → **Leptokurtic**

**10.9** Dataset A: excess kurtosis 1.5; Dataset B: −0.8. Which has fatter tails?

**A** (leptokurtic, excess > 0) has fatter tails / more outliers; B is platykurtic

**10.10 (Cyber)** Login-attempt data has kurtosis $\beta_2 = 7$. What does this suggest?

Excess = 4 > 0 → **Leptokurtic**
Rare-but-extreme events (possible attacks) happen more often than a Normal model predicts → monitor closely.